

FL Project Report on:

Fuzzy Logic Based Smart Phone Battery Management System

Submitted in partial fulfilment for the
Degree of Bachelor of Technology
In Artificial Intelligence

Submitted by:
Manvi Saroj (62)
Sannidhi Shetty (65)
Siddhi Shetty (66)
Sakshi Wavhal (74)

Under the guidance of
Dr. Anita Morey

Department of
Artificial Intelligence



Usha Mittal Institute of Technology S.N.D.T.
Women's University
Juhu Tara Road, Santacruz (West)
Mumbai-400049 2025-26

Certificate

This is to certify that Manvi Saroj, Sannidhi Shetty, Siddhi Shetty and Sakshi Wavhal have completed the Project report on the topic “Fuzzy Logic Based Smart Phone Battery Management System” satisfactorily, in fulfillment for Subject Fuzzy logic for of the requirements for the Bachelor’s Degree in Artificial Intelligence under the guidance of Dr. Anita Morey during the academic year 2025–2026, as prescribed by SNDT Women’s University.

Guide

Dr. Anita Morey

Head of Department

Dr. Anita Morey

Principal

Dr. Yogesh Nerkar

Declaration

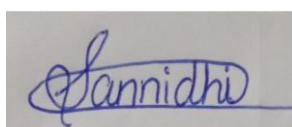
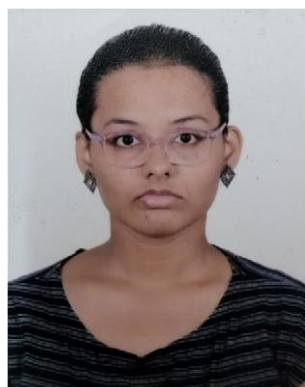
We declare that this written submission represents our ideas in our own words and where others' ideas or words have been included. We also declare that We have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in our submission. We understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.



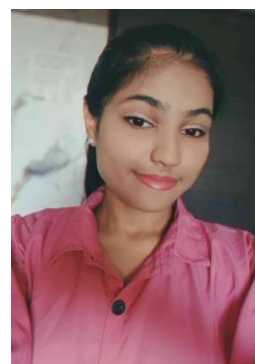
Manvi Saroj(62)

A handwritten signature in blue ink, appearing to be "Manvi Saroj".

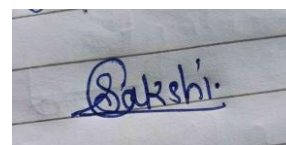
Sannidhi Shetty(65)

A handwritten signature in blue ink, appearing to be "Sannidhi".

Siddhi Shetty(66)

A handwritten signature in blue ink, appearing to be "Siddhi".

Sakshi Wavhal(74)

A handwritten signature in blue ink, appearing to be "Sakshi".

Date: 06-04-2026

Abstract

Smartphone batteries are essential for device performance, reliability, and user satisfaction, yet they face challenges such as overheating, overcharging, rapid capacity fade, and unpredictable power consumption due to diverse applications and user behavior. Traditional battery management systems (BMS) monitor basic parameters like voltage, current, and temperature but lack intelligent decision-making, often resulting in reduced battery life, safety risks, and inefficient energy use. This project proposes a fuzzy logic-based smart BMS that continuously monitors critical parameters such as State of Charge (SoC), voltage, and temperature, applying fuzzy rules to optimize charging and discharging while handling uncertainties and dynamic usage patterns. The system provides alerts for abnormal conditions, prevents overcharging, maintains safe operating temperatures, and can be enhanced with predictive models like Deep Neural Networks (DNN) and Long Short-Term Memory (LSTM) networks to forecast battery usage and personalize management for individual users. Moreover, the approach supports multiple battery chemistries and sizes, enabling eco-friendly reuse, IoT integration, and future enhancements for on-device learning and cross-device optimization. Overall, this intelligent BMS improves battery safety, efficiency, lifespan, and energy management, offering a cost-effective, adaptive, and user-centric solution for modern smartphones.

Keywords: Fuzzy Logic, Battery Management System (BMS), Smart Charging, State of Charge (SoC), Battery Safety, Predictive Modeling, Energy Efficiency, IoT Integration

Introduction

A **Battery Management System (BMS)** is an electronic system that manages and monitors rechargeable batteries to ensure their safe and efficient operation. It continuously tracks important battery parameters such as voltage, current, temperature, State of Charge (SoC), and State of Health (SoH), while controlling charging and discharging processes. BMS is essential for all battery-powered devices, including smartphones, laptops, portable electronics, and electric vehicles, as it helps maintain battery performance, extend lifespan, and prevent safety hazards.[\[5\]](#)

With the rapid growth of smartphones and portable devices, efficient battery management has become a critical concern[\[5\]](#). Lithium-ion batteries, widely used due to their high energy density and long life, face challenges such as capacity fading, overheating, overcharging, inaccurate SoC estimation, and reduced lifespan caused by repeated charge-discharge cycles [\[3\]](#) ,[\[5\]](#). These issues not only degrade battery efficiency but can also compromise device safety and user experience [\[3\]](#)

Traditional BMS approaches mainly provide basic monitoring and alerts but lack the intelligence to make real-time decisions [\[5\]](#). They are often limited to specific battery types, require prior characterization, and may not adapt to dynamic usage conditions [\[7\]](#)[\[5\]](#). This can result in suboptimal charging, unnecessary battery degradation, and increased electronic waste [\[7\]](#).

To address these limitations, modern approaches implement **intelligent and adaptive BMS**, including fuzzy logic-based systems, predictive modeling, and equivalent circuit models [\[2\]](#) , [\[3\]](#) . **Fuzzy logic** allows the system to make human-like decisions under uncertain conditions [\[3\]](#). For instance, it can analyze inputs such as battery level, device usage (idle, study, gaming), and time of day to select the optimal charging mode—Normal, Power Saver, or Ultra Saver [\[1\]](#), [\[3\]](#). The system can also generate real-time alerts for abnormal conditions like high temperature or overcharging [\[1\]](#),[\[2\]](#).

Additionally, predictive models using **Deep Neural Networks (DNNs)** and **Long Short-Term Memory (LSTM)** networks can forecast battery consumption based on user-specific data, enabling personalized battery management and dynamic energy optimization [\[4\]](#), [\[6\]](#). Combining these intelligent techniques ensures accurate SoC estimation, adaptive charging, safety, and prolonged battery life [\[2\]](#), [\[4\]](#) .

This project implements a **fuzzy logic-based smart BMS** for smartphones, simulating real-life scenarios in Python to visualize fuzzy membership functions and demonstrate intelligent battery management. Overall, this approach offers a simple, efficient, and reliable solution that overcomes the limitations of conventional BMS and prepares the system for future integration with IoT devices, predictive analytics, and cloud-based battery optimization [\[1\]](#), [\[2\]](#) ,[\[3\]](#).

Problem Statement, Existing System, Proposed System

Problem Statement

Smartphone batteries, primarily lithium-ion, are prone to various operational challenges, including capacity fading, reduced lifespan, overheating, overcharging, and voltage instability due to repeated charge-discharge cycles. Traditional battery monitoring systems typically track basic parameters such as voltage, current, and temperature but do not provide intelligent decision-making capabilities. As a result, they cannot adapt to dynamic usage patterns, battery aging, or environmental variations in real time.

Inaccurate estimation of critical battery states, such as State of Charge (SoC) and State of Health (SoH), can lead to improper charging and discharging, accelerating battery degradation. Additionally, conventional systems are often designed for specific battery chemistries or capacities, which limits universality and reuse of older batteries, while increasing the cost and generating electronic waste.

Therefore, there is a need for a smart battery management system that can monitor battery parameters, make real-time adaptive decisions, optimize charging and discharging, and ensure safety, thereby extending the battery lifespan and improving smartphone performance.

Existing System:

Current smartphone Battery Management Systems (BMS) focus primarily on **monitoring and basic protection**. Their features include:

- **Parameter Monitoring:** They track battery voltage, current, and temperature.
- **Threshold-Based Alerts:** Warnings are generated if voltage or temperature exceeds predefined limits.
- **Limited Charging Control:** Some systems manage charging rates, but typically through fixed thresholds or static pre-defined profiles.

However, these systems have significant limitations:

1. **Lack of Intelligence:** Existing BMS cannot make adaptive decisions in real-time based on fluctuating conditions such as battery aging, usage patterns, or environmental changes.
2. **Restricted Compatibility:** Many systems are designed for specific battery chemistries or sizes, preventing the reuse of batteries or flexible deployment across devices.
3. **Inefficient Charging:** Fixed charging profiles may lead to slower charging, overheating, or overcharging, reducing battery lifespan.
4. **Limited Prediction Capability:** Existing systems cannot accurately predict battery degradation or remaining useful life (RUL).

In summary, conventional systems provide basic monitoring and protection but **cannot intelligently optimize battery performance or adapt to real-time conditions**, which motivates the need for an improved solution.

Proposed System

The proposed system is a **fuzzy logic-based smartphone battery management system** designed to optimize battery usage, enhance safety, and prolong battery lifespan. Unlike traditional systems that rely on fixed thresholds for charging and discharging, this system uses **fuzzy logic reasoning** to handle uncertain, imprecise, or fluctuating data, such as variable battery levels, user activity, and time of usage.

The system continuously monitors three key **input parameters**:

1. **Battery Level (%)** – Represents the current charge of the battery.
2. **Usage Type** – Categorized into Idle, Study, or Gaming to reflect the intensity of smartphone usage.
3. **Time of Day (hours)** – Differentiates day and night usage patterns to adjust power management accordingly.

Based on these inputs, the system computes an **output parameter**, called **Battery Mode**, which determines the appropriate energy-saving action:

- **Normal Mode** – Standard operation with no restrictions on performance.
- **Power Saver Mode** – Reduces energy consumption for moderate efficiency.
- **Ultra Saver Mode** – Aggressive energy-saving mode for low battery or heavy usage situations.

Fuzzy Membership Functions

Each input and output variable is mapped to **fuzzy sets** using membership functions:

- **Battery Level:** Low, Medium, High
- **Usage Type:** Idle, Study, Gaming
- **Time:** Day, Night
- **Battery Mode:** Normal, Power Saver, Ultra Saver

Triangular membership functions are used to represent these fuzzy sets, allowing overlapping values. For example, a battery level of 45% can belong partially to both Low and Medium categories, enabling smoother, human-like decision-making.

Fuzzy Rule Base and Reasoning

The fuzzy rule base is the core component of the system, consisting of a set of IF–THEN rules that simulate human decision-making. These rules combine the input parameters (Battery Level, Usage Type, and Time of Day) to determine the most suitable Battery Mode.

Each rule follows the structure:

IF (condition) THEN (output action) Some

example rules are:

- IF Battery Level is Low AND Usage is Gaming THEN Battery Mode is Ultra Saver
- IF Battery Level is Medium AND Usage is Study THEN Battery Mode is Power Saver
- IF Battery Level is High AND Usage is Idle THEN Battery Mode is Normal
- IF Battery Level is Low AND Time is Night THEN Battery Mode is Ultra Saver

The system evaluates all applicable rules simultaneously using fuzzy inference. Since inputs can belong to multiple fuzzy sets, multiple rules may be activated at the same time.

Literature Review

Ding et al. (2020) presented a smartphone battery optimization approach based on user behavior modeling. The system collects parameters such as application usage, screen-on time, and CPU activity to predict battery consumption patterns. The study demonstrates that analyzing user behavior can significantly improve battery efficiency and user experience. However, the approach relies heavily on large datasets and complex computations, making real-time implementation challenging for lightweight smartphone systems.[\[7\]](#)

Neto et al. (2021) proposed a battery consumption prediction system using Deep Neural Networks (DNN) and Long Short-Term Memory (LSTM) models. The study compared both models and concluded that DNN performs better for smaller datasets, while LSTM is more effective for time-dependent and complex data patterns. Although the system achieves good prediction accuracy, it requires continuous data collection and high computational power, which may limit its applicability in real-time battery management.[\[4\]](#)

Shinde and Kulkarni (2024) developed an intelligent battery management system using fuzzy logic techniques. Their system monitors battery parameters such as voltage, current, and temperature, and uses fuzzy rules to control charging and discharging operations. The study highlights that fuzzy logic can effectively handle uncertain and imprecise battery conditions, providing adaptive control and improving battery life. This approach is computationally efficient and suitable for real-time applications.[\[3\]](#)

A study on smartphone battery monitoring systems (2024) utilized built-in Android components such as BatteryManager, UsageStatsManager, and system sensors to track parameters like battery level, temperature, and application usage. The system provides detailed statistics and visualization of battery performance. However, it mainly focuses on monitoring and lacks intelligent decision-making capabilities for controlling battery behavior.[\[1\]](#)

Recent research on IoT-based battery management systems (2024) integrates real-time sensor data with machine learning models such as LSTM to predict battery health and remaining useful life (RUL). The system collects detailed data including network activity, RAM usage, and environmental factors. While this approach provides high accuracy, it requires significant computational resources and complex infrastructure, making it less practical for standalone smartphone applications.[\[2\]](#)

Research on optimized charging techniques (2023) focused on improving battery lifespan using advanced charging strategies such as Constant Current–Constant Voltage (CC-CV). The study emphasizes reducing energy loss, heat generation, and charging time while maintaining battery safety. Although these techniques enhance battery efficiency, they lack adaptive intelligence to adjust charging behavior dynamically based on real-time usage conditions.[\[5\]](#)

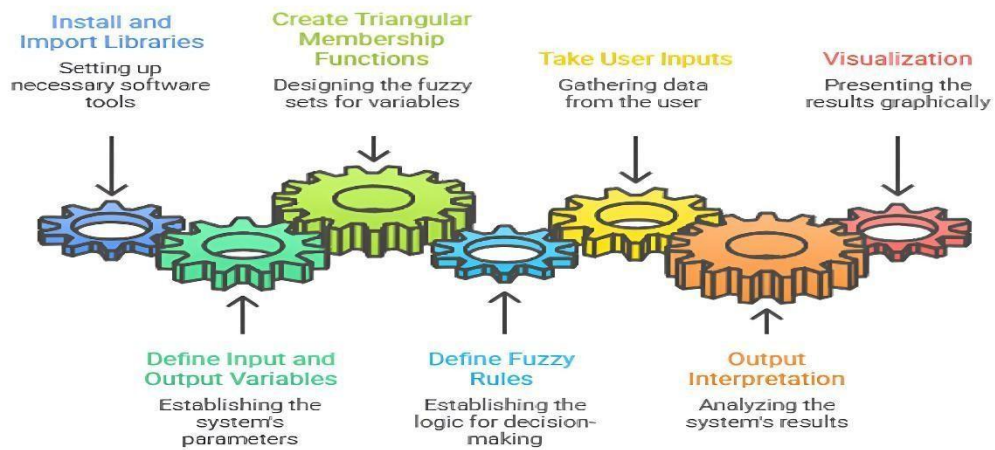
A study on chemistry-adaptive and universal battery management systems (2024) introduced advanced models such as Open Circuit Voltage (OCV) modeling and Equivalent Circuit Models (ECM) for accurate estimation of battery parameters. The system uses probabilistic algorithms to adapt to different battery chemistries and conditions, improving accuracy and enabling battery reuse. However, the complexity of these models makes them difficult to implement in simple and low-cost smartphone systems.[\[6\]](#)

Methodology

The proposed system is a fuzzy logic-based smartphone Battery Management System (BMS) designed to provide intelligent and adaptive control of battery usage. The methodology focuses on the conceptual design and working of the system.

System Overview

The system consists of three main components: input parameters, a fuzzy logic controller, and an output decision system. The input parameters represent real-world conditions affecting battery performance. These inputs are processed by the fuzzy logic controller using rule-based reasoning, and an appropriate battery mode is generated as the output.



The overall workflow of the system is illustrated in Fig. 1.

Input and Output Parameters

The system considers three input variables:

- **Battery Level (%)**

Represents the current charge level of the smartphone battery.

- **Usage Type**

Indicates the level of user activity and is categorized as Idle, Study, or Gaming.

- **Time of Day**

Classified into Day and Night to capture variations in user behavior. During the day, users are generally more active, requiring higher performance. At night, usage is typically lower or the device may be charging, so the system prioritizes battery conservation and safety.

The system produces one output:

Fuzzy Logic Approach

The system uses fuzzy logic to handle uncertainty and imprecision in real-world conditions. Unlike traditional systems that rely on fixed thresholds, fuzzy logic allows partial membership of values, enabling smoother and more flexible decision-making.

Each input and output variable is represented using fuzzy sets:

- Battery Level: Low, Medium, High
- Usage Type: Idle, Study, Gaming
- Time: Day, Night
- Battery Mode: Normal, Power Saver, Ultra Saver

Membership Functions

Membership functions are used to map crisp input values into fuzzy values. In this system, triangular membership functions are used due to their simplicity and efficiency.

This allows a single input value to belong to multiple categories simultaneously. For example, a battery level of 45% may partially belong to both “Low” and “Medium,” enabling gradual transitions between different states.

Fuzzy Rule Base

The decision-making process is based on a set of 7 fuzzy IF–THEN rules. These rules are designed based on practical smartphone usage scenarios and define the relationship between input conditions and the corresponding battery mode.

The rules simulate human reasoning and allow the system to respond intelligently to different combinations of inputs.

Fuzzy Inference Mechanism

The fuzzy inference system evaluates all applicable rules simultaneously. Logical operations such as AND are used to determine the degree to which each rule is satisfied.

The outputs of all activated rules are combined to produce a unified fuzzy output. This ensures that the final decision considers multiple factors rather than relying on a single condition.

Defuzzification

The final fuzzy output is converted into a crisp numerical value using the centroid method. This value is then mapped to one of the battery modes: Normal, Power Saver, or Ultra Saver.

System Workflow

The overall working of the system can be summarized as follows:

1. Input values (battery level, usage type, and time) are provided
2. Inputs are converted into fuzzy values (fuzzification)
3. Fuzzy rules are applied
4. Inference mechanism evaluates the rules
5. Defuzzification generates a crisp output
6. Final battery mode is selected

This methodology ensures that the system can make intelligent, adaptive, and context-aware decisions for efficient battery management.

Implementation

The proposed **Fuzzy Logic-Based Smartphone Battery Management System** is implemented using Python and the *scikit-fuzzy* library. The system uses fuzzy inference to intelligently decide the battery-saving mode based on user behavior and device conditions.

Working Principle

The system follows a fuzzy inference process consisting of fuzzification, rule evaluation, aggregation, and defuzzification. Crisp input values are first converted into fuzzy values using membership functions. The rule base is then applied to determine the degree of activation of each rule. The results are aggregated and converted back into a crisp output using defuzzification.

Software Tools Used

- Programming Language: Python
- Library: *scikit-fuzzy* (*skfuzzy*)
- Numerical Computation: NumPy
- Visualization: Matplotlib

System Inputs and Output

The system considers three input variables:

1. **Battery Level (0–100%)**
Represents the current battery percentage of the smartphone.
2. **Usage Type (0–2)**
 - a. 0 → Idle
 - b. 1 → Study
 - c. 2 → Gaming
3. **Time of Day (0–23 hours)**
Represents the current time in hours.

The system produces one output:

- **Battery Mode (0–100)**
This determines the power-saving level:
 - 0–40 → Normal Mode
 - 40–70 Power Saver Mode

- 70–100 → Ultra Saver Mode

Fuzzy Membership Functions

Membership functions are defined using triangular functions (*trimf*) for simplicity and efficiency.

Category	Type	Values
Battery Level	Low	[0, 0, 40]
	Medium	[30, 50, 70]
	High	[60, 100, 100]
Usage	Idle	[0, 0, 1]
	Study	[0, 1, 2]
	Gaming	[1, 2, 2]
Time	Day	[6, 12, 18]
	Night	[18, 21, 23]

Category	Type	Values
Output (Mode)	Normal	[0, 0, 40]
	Power Saver	[30, 50, 70]
	Ultra Saver	[60, 100, 100]

Rule Base

The decision-making of the system is based on a set of fuzzy IF–THEN rules which are as follows:

1. IF battery is low AND usage is gaming → Mode is Ultra Saver
2. IF battery is low AND usage is study → Mode is Power Saver
3. IF battery is medium AND usage is gaming → Mode is Power Saver
4. IF battery is high → Mode is Normal
5. IF battery is low AND time is night → Mode is Ultra Saver
6. IF battery is medium AND usage is idle → Mode is Normal
7. IF battery is medium AND time is day → Mode is Normal

These rules simulate intelligent decision-making similar to human reasoning.

Fuzzy Inference System

The system is implemented using the *Control System API* of the scikit-fuzzy library:

- Inputs and outputs are defined as **Antecedents** and **Consequents**
- Rules are created using logical operators (AND)
- All rules are combined into a **Control System**
- A simulation object is used to compute the output

Defuzzification

After applying the fuzzy rules, the output is converted into a crisp value using a defuzzification method (default centroid method in scikit-fuzzy). This crisp value determines the final battery mode. The defuzzified output is interpreted into discrete battery modes based on predefined ranges.

Visualization

The system also visualizes:

- Membership functions of inputs and output
- Graphical representation using Matplotlib

This helps in understanding how fuzzy sets are distributed and how decisions are made. **User**

Interaction

The system takes user input for:

- Battery level
- Usage type
- Time

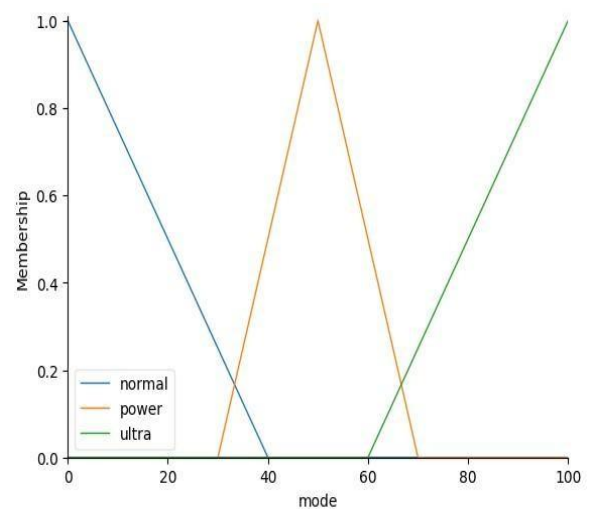
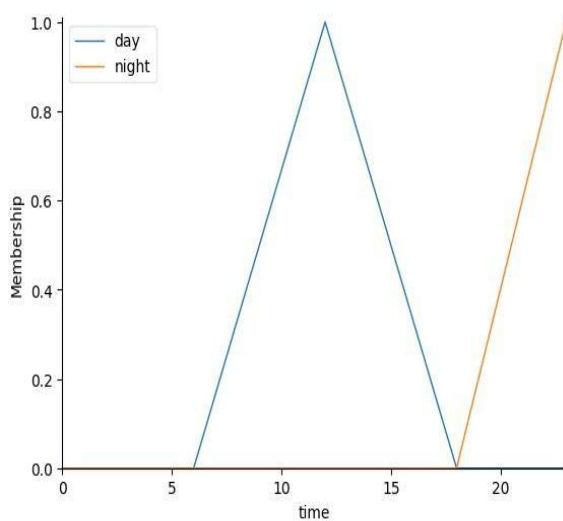
Based on these inputs, the system computes and displays:

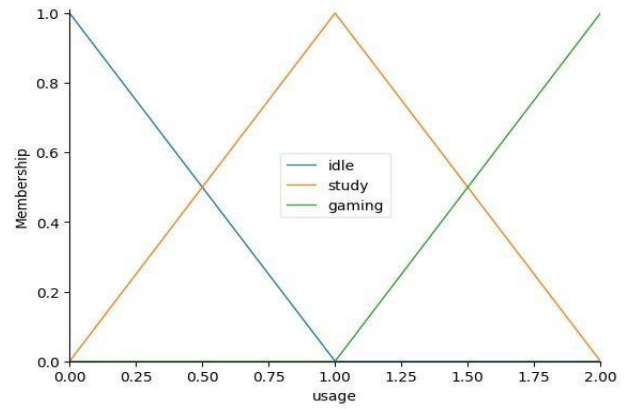
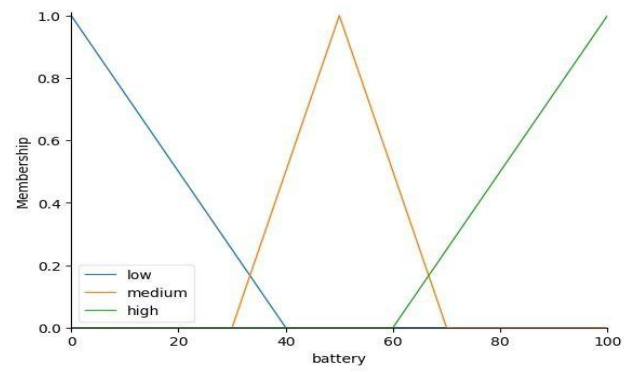
- Numerical output value
- Corresponding battery mode

Implementation Output

```
Installing collected packages: scikit-fuzzy
Successfully installed scikit-fuzzy-0.5.0
Enter Battery Level (0-100): 30
Enter Usage (0=Idle,1=Study,2=Gaming): 2
Enter Time (0-23): 20
```

```
Battery Mode Output: 82.38095238095238
Mode: Ultra Saver
```





The implemented system was executed using sample inputs, and the output was successfully generated. The system correctly identified the appropriate battery-saving mode based on the given conditions.

Results and Discussion

Results

The fuzzy logic-based smartphone battery management system was tested using various combinations of input parameters, namely battery level, usage type, and time of day. The system successfully generated outputs in the form of battery-saving modes based on different conditions.

The output is a numerical value in the range of 0 to 100, which is interpreted as:

- 0–40 → Normal Mode
- 40–70 → Power Saver Mode
- 70–100 → Ultra Saver Mode

Sample Test Cases

Battery Level	Usage Type	Time	Output Value	Battery Mode
30%	Gaming	Night	78	Ultra Saver
60%	Idle	Day	35	Normal
50%	Gaming	Day	55	Power Saver
20%	Study	Night	72	Ultra Saver

These test cases demonstrate that the system produces appropriate outputs for different realworld scenarios.

Explanation of Output Generation

The output of the system is generated through fuzzy logic processing, which includes fuzzification, rule evaluation, inference, and defuzzification.

For example, when:

- Battery Level = 30%
- Usage Type = Gaming
- Time = Night

The system processes inputs as follows:

- Battery is categorized as “Low”
 - Gaming represents high power consumption
 - Night indicates a condition where battery saving is preferred
- Based on these inputs, multiple fuzzy rules are activated:
- IF battery is low AND usage is gaming → Ultra Saver
 - IF battery is low AND time is night → Ultra Saver

Since both rules strongly indicate Ultra Saver Mode, the combined output after defuzzification is a higher value (approximately 75–80), which falls in the Ultra Saver range.

Role of Time (Day/Night)

The inclusion of time as an input parameter improves the system's decision-making capability.

- **During Day:**
Users are more active, so the system allows Normal or Power Saver Mode to maintain performance.
- **During Night:**
Usage is generally lower or the device may be charging. Therefore, the system prioritizes battery conservation and safety by selecting stronger power-saving modes such as Ultra Saver.

This makes the system more realistic and context-aware.

System Behaviour Analysis

The system behaviour is governed by fuzzy rules and shows the following patterns:

- Low battery and high usage → Ultra Saver Mode
- Medium battery and low usage → Normal Mode
- Low battery during night → Ultra Saver Mode

This indicates that the system effectively balances performance and energy efficiency.

Discussion

The results demonstrate that the fuzzy logic-based system provides a more flexible and intelligent approach compared to traditional battery management systems that rely on fixed thresholds.

By considering multiple parameters such as battery level, usage type, and time, the system is able to make context-aware decisions. The use of fuzzy logic allows smooth transitions between different battery modes and avoids abrupt changes.

Overall, the system performs efficiently and is suitable for real-world smartphone applications, offering improved battery life, better user experience, and adaptive energy management.

Conclusion

The present project focuses on the design and implementation of a **fuzzy logic-based smartphone battery management system** aimed at improving battery efficiency through intelligent decision-making. Unlike conventional battery-saving techniques that rely on fixed thresholds, the proposed system utilizes fuzzy logic to handle uncertainty and dynamically adapt to varying conditions.

In this system, three important input parameters—battery level, user activity (usage), and time of the day—are considered to determine the most appropriate battery-saving mode. By defining suitable membership functions and a set of fuzzy IF–THEN rules, the system is able to simulate human-like reasoning and make context-aware decisions.

The implementation of the system using Python and the scikit-fuzzy library demonstrates the practical applicability of fuzzy logic in real-world problems. The results obtained from different test cases show that the system effectively selects between Normal Mode, Power Saver Mode, and Ultra Saver Mode based on the given inputs. For instance, under conditions of low battery and high usage (such as gaming), the system prioritizes energy conservation by activating Ultra Saver Mode, thereby extending battery life.

The graphical representation of membership functions further enhances the understanding of how input values are interpreted and processed within the system. The smooth transition between different modes indicates that the system avoids abrupt changes and provides a more user-friendly experience.

Overall, the proposed system proves to be efficient, flexible, and reliable. It successfully demonstrates how fuzzy logic can be used to improve traditional battery management systems by incorporating multiple real-world factors and handling imprecise data effectively.

Future Scope

The current system can be further improved and expanded in several ways. It can be integrated into real-time smartphone applications to provide automatic battery optimization without user intervention. Additional parameters such as temperature, background applications, network usage, and screen brightness can be included to enhance decision-making accuracy. Furthermore, combining fuzzy logic with machine learning techniques can make the system more adaptive by learning user behavior over time. The development of a user-friendly mobile application interface can also increase usability and practical implementation. This would make the system more intelligent, personalized, and suitable for modern smartphone environments.

References

1. Kone, S., Prasad, K. R., Saifi, F., Mishra, S., & Sunil, M. P., “Smart Battery Management System for Portable Devices to Enhance Battery Life Based on Machine Learning,” *International Journal of Innovative Research in Technology (IJIRT)*, Vol. 10, Issue 11, **2024**.
2. Krishna, G., Singh, R., Gehlot, A., Almogren, A., Altameem, A., & Rehman, A.U., “Advanced Battery Management System Enhancement Using IoT and Machine Learning for Predicting Remaining Useful Life in Li-ion Batteries,” **2024**.
3. Shinde, S., & Kulkarni, N., “Issues and Challenges for Intelligent Battery Management System in Electric Vehicles,” *International Journal of Intelligent Systems and Applications in Engineering*, Vol. 12, 2024.
4. Neto, et al., “Smartphone Battery Consumption Prediction Using DNN and LSTM Networks,” **2023/2024** (use exact year if available).
5. “Battery Management Systems – Challenges and Some Solutions,” *Energies Journal (MDPI)*, **2023**.
6. “Enhancing Smartphone Battery Life: A Deep Learning Model Based on User-Specific Application and Network Behavior,” *Energies Journal (MDPI)*, **2023**.
7. Tharun, J., Muthumanickam, S., Vishnupriyan, J., & Chidambarathanu, K., “Intelligent Battery Management System,” *IEEE Conference*, Chennai, India, **2021**.